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ROOT CAUSE UNLOCKED

Unlock Your *GLP-1*

Keep the Weight Loss. Keep Your Strength, Your Skin, and Your Long-Term Health, Too.

A guide for adults losing weight on semaglutide or tirzepatide (Ozempic, Wegovy, Mounjaro, Zepbound) who want to understand, and address, the cellular side of rapid fat loss that almost nobody explains at the pharmacy counter.

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

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This edition is pending Dr. Seay's clinical sign-off. Every citation has been verified against its live PubMed record, including a check for retractions. The companion citation ledger lists each source and the claims that were investigated but left uncited.

BEFORE YOU START

> ***Disclaimer:*** *This book is for educational purposes only and does not constitute medical advice. It does not diagnose any condition, and reading it does not make you my patient. Always consult a qualified healthcare provider before starting any new treatment, supplement, or dietary change, and do not start, stop, or change any prescribed medication without speaking to the prescriber. Keep taking your GLP-1 medication exactly as prescribed. If you have had persistent vomiting or months of very low intake and then develop confusion, unsteadiness, or vision changes, treat that as an emergency and read Key 5 now.*

A note before you start. This guide is educational information. It explains the biology behind a documented consequence of rapid GLP-1 driven weight loss and the general categories of intervention used to address it. It is not individualized medical advice, and it does not replace a relationship with your prescribing physician or a clinical evaluation with my team. Keep taking your GLP-1 medication exactly as prescribed. Talk to your doctor or a qualified practitioner before starting any new supplement, especially if you take blood thinners, thyroid medication, diuretics, or other prescriptions.

Contents

INTRODUCTION

What You Are About to Learn

If you are losing weight on a GLP-1 medication, it is working. The appetite reduction is real, the number on the scale is moving, and the metabolic benefits are genuine and well documented. That is not in question.

Here is what usually is not covered in a fifteen-minute prescribing appointment: the speed of that weight loss creates a cellular consequence that has nothing to do with your willpower, your diet, or your age, and everything to do with how fast your body is mobilizing fat. It is a mechanism, not a mystery, and it is documented in peer-reviewed research. Once you understand it, you can do something about it while you keep taking your medication exactly as prescribed.

This guide exists so you can keep the result you came for, the weight loss, without quietly trading away the muscle, the skin elasticity, the energy, and the long-term cellular health that most people do not realize are on the table. The goal is not to talk you out of your medication. The goal is that you go into this with your eyes open, protecting your strength and your energy on the way down rather than discovering a year from now that you lost more than fat. I am not going to tell you how much better you will feel or by when. What I can tell you is what is worth protecting and how.

Below are eight keys. Each one explains a specific consequence of rapid GLP-1 driven fat loss, the mechanism behind it in plain language, and what you can do about it at an educational level. The last section walks through the structured 90-day program I built to address all of this directly, if you decide you want a guided, lab-directed approach rather than piecing it together yourself.

KEY 1

The Weight Is Coming Off. So Is Something Else.

The problem nobody mentions. Your GLP-1 medication is doing exactly what it was designed to do. But when fat tissue breaks down as fast as it does on these medications, it releases fatty acids into your bloodstream faster than your mitochondria can process them. That overload has a name and a documented downstream effect that your prescriber's monitoring panel is not built to catch.

The mechanism, in plain terms. GLP-1 receptors do not only sit on the pancreas and the appetite centers in your brain, the places your medication is meant to act. They are also present on adipose-derived stem cells and the fibroblasts in your skin.[1] When your medication stimulates those receptors, it reduces the protective cytokines those stem cells normally produce to shield nearby fibroblasts from oxidative damage.[2] At the same time, the fatty acid flood from rapid fat mobilization generates a surge of reactive oxygen species, unstable molecules that damage DNA, break down collagen, and overwhelm your natural antioxidant defenses. This combination is documented in a 2025 peer-reviewed paper on GLP-1 receptor agonists and skin aging,[1] and separately in 2026 research on the metabolic and redox effects of GLP-1 therapy.[3]

Cells that absorb too much of this oxidative damage do not simply die and get cleared away. They enter a state called senescence: they stop functioning, stop dividing, but refuse to die. Researchers call them zombie cells. Instead of going quietly, senescent cells release a continuous stream of inflammatory signaling molecules known as SASP, the Senescence-Associated Secretory Phenotype.[4]

Where the certainty stops, stated plainly. The senescence and SASP biology above is well established in the laboratory. The step from that biology to the specific symptoms in the next two keys, in a specific person on a GLP-1, is a proposed mechanism rather than a proven chain. Nobody has run the trial that would settle it. I use this model clinically because it fits what I see and because everything it points to (protein, training, nutrient repletion, clearance) is low risk and defensible on its own terms. You should know which of those two things you are buying.

What to do about it, at an educational level. This is not a reason to stop your medication but a reason to add a second, complementary layer of support that addresses the cellular side effect directly: clearing the senescent cells that have already formed, rebuilding the antioxidant reserves that were depleted, and giving your body the structural building blocks (protein, collagen, lymphatic support) it needs to keep pace with how fast you are losing weight. The rest of this guide walks through each of those pieces.

KEY 2

Up to Half the Weight You Lose Might Be Muscle, Not Fat

The half of weight loss nobody warns you about. Most people assume weight loss on a GLP-1 medication is fat loss. It is not entirely.[5] A landmark study on semaglutide published in Cell Metabolism found that in the STEP-1 trial, lean mass accounted for roughly 45 percent of total weight lost, nearly double the rate that would be expected from caloric restriction alone. That is muscle, and once it is gone, it does not come back just because you stopped losing weight.

Why muscle goes, in plain language. The muscle loss associated with GLP-1 therapy is not caused by one single thing. Research reviewing the pattern across GLP-1 patients points to several factors working together: reduced protein intake because appetite suppression makes it easy to under-eat protein specifically, a negative energy balance that pushes your body toward breaking down tissue for fuel, hormonal shifts, impaired signaling through the mTOR pathway that normally tells your body to build and maintain muscle, and increased protein breakdown.[6] For patients on tirzepatide specifically, there is additional evidence that the GIP component of the medication can push muscle precursor cells toward becoming fat cells instead of muscle, compounding the effect.

The consequence is not cosmetic. Reduced grip strength, functional decline, and accelerated age-related muscle loss (sarcopenia) beyond what caloric restriction alone would explain have all been documented in GLP-1 treated patients, including in a 24-month retrospective study specifically measuring muscle mass, grip strength, and gait speed.[7]

The two levers that matter, at an educational level. Two things counter this mechanism directly, and they are the two most important actions in this entire guide: protein and resistance training. Your target is 1.6 to 2.0 grams of protein per kilogram of body weight per day, distributed across three to four meals rather than back-loaded into one. Because appetite suppression makes this hard, prioritize protein first at every meal, before anything else. Pair that with resistance training a minimum of three times per week, using compound movements (squats, rows, presses, hip hinges) worked to within two to three repetitions of failure, with a protein-rich meal or shake within two hours after each session.[8] This is the single most direct lever you have against the lean mass consequence of rapid GLP-1 driven weight loss, and it also independently activates a cellular cleanup process called autophagy, which works alongside the senolytic support described in Key 4.

A third, smaller lever: omega-3 fats. After protein and resistance training, one more input has real human trial evidence, at a much more modest level. A 2023 meta-analysis of randomized trials found that omega-3 supplementation produced a very small but statistically significant benefit to muscle strength, with no meaningful effect on muscle mass or function overall.[25] The most encouraging single trial was run in exactly the group with the most to lose: healthy older adults, aged 60 to 85, who took about 3.4 grams of combined EPA and DHA daily for six months and gained 3.6 percent thigh muscle volume, 2.3 kilograms of grip strength, and 4 percent on one-rep-max strength compared with the control group.[26] Keep the size of this in perspective. Omega-3 is a garnish, not the meal, and nothing about it replaces the protein target or the training. But if you are an older adult on a GLP-1, or you already take fish oil for other reasons, the evidence suggests it is quietly working in your favor. The doses studied were in the low single-digit grams of combined EPA and DHA per day, which is considerably more than a standard one-capsule fish oil provides, so check your label. And as with everything in this guide, clear any new supplement with your prescriber first, especially if you take a blood thinner.

KEY 3

Why Your Skin, Joints, and Energy Feel Off, and Your Prescriber Cannot Explain It

The symptoms your monitoring panel will not catch. Accelerated skin aging or looseness disproportionate to the weight lost, sometimes called "Ozempic face." New joint pain or diffuse achiness that started after you began your medication. Fatigue that does not track with your improving blood sugar or cholesterol numbers. Brain fog, word-finding trouble, or memory changes. Increased hair shedding. These symptoms typically show up somewhere between three and twelve months into GLP-1 therapy, and they are frequently written off as aging, inadequate nutrition, or "just how weight loss feels." Standard GLP-1 monitoring panels, the ones checking blood sugar and lipids, are not built to catch what is actually driving them.

One proposed source behind all of it. The proposed common source for this whole cluster is the senescent, or "zombie," cells described in Key 1 and the SASP inflammatory signal they release continuously. Laboratory work indicates that SASP cytokines degrade the extracellular matrix that supports your skin and joints, which is the proposed mechanism behind both the accelerated skin aging and the new joint discomfort. SASP also induces an enzyme called CD38 in neighboring healthy cells, which depletes their NAD+, a molecule your cells depend on for energy production.[9] That is a plausible contributor to the fatigue and brain fog that do not improve even as your metabolic numbers do, and I want to be clear that plausible is not the same as demonstrated in patients like you. The hair changes trace back to the same cellular stress and nutritional depletion that comes with rapid weight loss on a reduced-appetite diet.

A different way to read these symptoms, at an educational level. The important reframe here is that this symptom cluster is not an inevitable cost of GLP-1 therapy, and it is not "just aging." There is a specific and plausible process behind it, and every lever it points to is addressable. It responds to the same interventions covered in the rest of this guide: Key 4 covers clearing the senescent cells generating the SASP signal, Key 5 covers rebuilding the antioxidant and NAD+ reserves that were depleted, and Key 6 covers supporting the lymphatic clearance pathways that remove the inflammatory debris those cells leave behind. If any of these symptoms are severe or worsening rather than mild and stable, that is worth a direct conversation with your prescribing physician, not just a supplement.

KEY 4

Clearing the Zombie Cells

A burden that only grows. Senescent cells do not resolve on their own. Once a cell enters senescence, it stays there, continuing to release inflammatory SASP signals indefinitely, and it can even nudge nearby healthy cells toward senescence too. The longer they accumulate, the larger the inflammatory burden becomes. This is why the symptoms in Key 3 tend to build gradually rather than appear and resolve on their own.

How senolytics work, simply put. A category of natural compounds called senolytics has been studied specifically for their ability to selectively clear senescent cells while leaving healthy cells alone. In a screening of candidate compounds conducted at the Mayo Clinic, fisetin, a flavonoid found in strawberries and other fruits, was identified as the most potent natural senolytic tested.[10] Quercetin and curcumin are studied alongside it for complementary antioxidant and anti-inflammatory effects on the same pathway.

One detail matters clinically: senolytics do not work like a typical daily supplement. Research on senolytic mechanisms describes a “hit-and-run” effect, meaning the compounds do their work in a short window and then can be stopped, rather than needing continuous daily exposure to be effective.[11] This is why senolytic protocols are typically built around short pulsed cycles rather than everyday dosing.

What you are buying here, stated plainly. The senolytic evidence is laboratory work, animal work, and a small number of early human studies. There are no human outcome trials showing that fisetin, quercetin, or curcumin improve how a GLP-1 patient feels or functions, and I could not find one. That is what happens to compounds nobody can patent: the trials never get funded, and the marketing fills the gap. So what this section describes is reasoned clinical practice held to an endpoint, not trial-proven certainty. Two consequences worth acting on. Put your money into the protein and the resistance training from Key 2 first, because those are the levers with real human evidence behind them. And hold anything you add here to a defined endpoint rather than running it indefinitely.

Using this wisely, at an educational level. If you are exploring senolytic support, understand that timing matters as much as the compound itself. Continuous daily high-dose use is not how these compounds were studied, and pulsed cycling is what the mechanism research points toward. This is also why senolytic support is best paired with the antioxidant and lymphatic support covered in the next two keys: the proposed model has clearing cells release their stored inflammatory contents one final time, and your body needs the capacity to process and clear that debris.

KEY 5

Refilling the Tank, and Why Guessing at Supplements Wastes Money

A double depletion that gets half the attention. Two things are draining at once, and only one of them gets talked about. The reactive oxygen species overload described in Key 1 draws down your antioxidant reserves, specifically glutathione and CoQ10, faster than diet alone typically replaces them. But there is a second, simpler drain that matters more and gets mentioned less: you are eating dramatically less food. In the trial that worked out how semaglutide actually produces weight loss, total energy intake across a day fell by 24 percent.[14] Your requirement for vitamins and minerals did not fall by 24 percent. Reviews of GLP-1 therapy now describe a recognizable pattern of micronutrient shortfalls that follow from sustained low intake, and they are not evenly distributed. Not every patient is short in the same places, or to the same degree.[13]

What is actually running low. Glutathione is your body's primary intracellular antioxidant, and GGT (an enzyme measurable on routine bloodwork) rises when glutathione turnover is under active oxidative stress. GLP-1 therapy also reduces circulating cholesterol and alters lipid flux in ways that are biologically linked to reduced CoQ10 and glutathione peroxidase levels, the mitochondrial cofactors your cells need for energy production. And as covered in Key 3, the SASP signal from senescent cells actively depletes NAD+ in surrounding healthy cells through CD38 induction.[9]

Thiamine, vitamin B1, deserves its own paragraph, because it behaves differently from the others. Your body holds only a small reserve of it relative to what it uses daily, so thiamine status falls within weeks of poor intake rather than over months. That timeline lines up uncomfortably well with GLP-1 therapy, particularly if nausea or vomiting has been part of your experience. This is not hypothetical for people who look like you: in one obesity medicine clinic population, about a third of patients already tested thiamine deficient before any surgery or medication.[17] Low thiamine is also common in type 2 diabetes.[18] Bariatric surgery creates the same combination of sharply reduced intake plus vomiting, and precisely because of that, it has an established thiamine monitoring protocol built around it.[16] Reviews are now asking directly whether GLP-1 care should borrow that framework rather than reinvent it.[15]

Seek care now, not later: the one emergency in this guide. If you have had persistent vomiting or months of very low intake on a GLP-1 medication, and you then develop confusion or unusual mental fuzziness, unsteadiness on your feet, or vision changes such as double vision or trouble moving your eyes, that combination is a medical emergency and not a supplement problem. It describes Wernicke's encephalopathy, which is caused by thiamine deficiency. Go to an emergency department the same day and say the words "possible thiamine deficiency," because treatment is intravenous thiamine and it needs to happen before the damage sets.

Keep this in proportion. It is rare. A systematic review of published cases found six reported with semaglutide,[19] and an analysis of the FDA adverse event database identified fifteen cases across GLP-1 medications as a whole.[20] Set against many millions of prescriptions, that is a very small number, and it is not a reason to stop your medication. But two details from those reports are worth carrying with you. Nearly every patient had prolonged gastrointestinal symptoms or substantial rapid weight loss first, meaning there was a warning period. And among patients followed afterward, lasting neurological problems were common, which is what makes early recognition matter so much. Caught early it is treatable. Missed, it may not be reversible.

The lab-first approach, at an educational level. The most useful mindset here is one I apply clinically: the most expensive supplement is the one you do not actually need. Rather than assembling a long list of antioxidant products by guesswork, the more precise approach is to find out, through bloodwork, what is actually depleted in your body and direct support specifically there. Thiamine is measurable, which puts it firmly in the "test, do not guess" category alongside the rest. This is exactly what lab-directed protocols are built to do, and it is a core piece of the structured program described at the end of this guide. If you are not working with a practitioner yet, the general categories worth knowing about are a glutathione precursor (NAC) paired with direct glutathione support, a CoQ10 source for mitochondrial support, and thiamine if your intake has been low for a sustained stretch or if nausea and vomiting have been a recurring part of your experience.

A word about NAD+ boosters, since they are marketed heavily to you. The underlying biology is real: senescent cells do deplete NAD+ in the tissue around them, which is why NMN and similar products are pitched to GLP-1 patients for fatigue and low cellular energy.[9] The human evidence is thinner than the marketing suggests. The trials that exist are small and measure laboratory markers rather than outcomes people can feel or that predict health years later.[24] More relevant to you specifically, a 2024 analysis linked the terminal breakdown products of excess niacin to major adverse cardiac events across multiple patient cohorts, with laboratory work showing one of those metabolites directly triggers blood vessel inflammation.[23] That is an association plus a plausible mechanism. It is not proof that taking an NAD+ precursor harms anyone, and it should not be reported to you as though it were. But niacin, NMN, and related compounds are cleared through the same pathway, and if you are on a GLP-1 medication you are, almost by definition, someone whose cardiovascular risk is actively being managed. An unresolved question about blood vessel inflammation, set against a benefit that has not been demonstrated, is a poor trade in your situation. This guide does not recommend routine NAD+ precursor supplementation.

If you do supplement thiamine, which form. Benfotiamine is a fat-soluble thiamine derivative that raises thiamine levels in the body more effectively than standard thiamine hydrochloride.[21] Better absorption is not the same as better results. The longest trial run to date gave benfotiamine for twelve months to people with type 2 diabetes and symptomatic nerve pain. It reliably raised every thiamine marker measured and was well tolerated, but it produced no significant improvement in nerve structure, nerve function, or symptoms compared with placebo.[22] Read honestly, that means benfotiamine is a legitimate option on absorption grounds and a poor bet as a treatment for established neuropathy. For the purpose that matters here, restoring a nutrient you may genuinely be short on, plain thiamine is the established and far less expensive choice.

KEY 6

Clearing the Debris, Not Just Stopping the Damage

What happens to the debris. Clearing senescent cells and quenching oxidative stress is only half the job. Dying cells, cleared cellular debris, and inflammatory SASP byproducts have to actually leave your tissue. If that clearance pathway is sluggish, the debris lingers and can re-trigger the same inflammatory signaling you are trying to resolve.

Your body's drainage route, explained. Your lymphatic system is the clearance pathway responsible for removing this kind of cellular and inflammatory debris at the tissue level. In a structured cellular restoration protocol, lymphatic support is designed to run continuously across the full course of treatment, specifically because it is the pathway responsible for clearing what the senolytic and antioxidant work leaves behind.

Keeping the pathway moving, at an educational level. This is the piece most people skip entirely because it is the least visible. Practical, low-cost support includes adequate hydration (a minimum of eight to ten glasses of filtered water daily), movement (even walking supports lymphatic flow, since unlike blood, lymph has no dedicated pump and relies on muscle contraction), and, if you are pursuing a structured protocol, dedicated lymphatic drainage support sequenced across the full course of your program rather than a single short course.

KEY 7

What to Actually Expect, Week by Week

Why the first weeks can mislead you. People starting a cellular restoration protocol alongside their GLP-1 medication often expect linear improvement from day one. It does not work that way, and not knowing this in advance is one of the most common reasons people quit a protocol that was, in fact, working exactly as expected.

The timeline, explained plainly. The model says that when senolytic support begins clearing senescent cells, those cells release their stored inflammatory contents one final time as they are eliminated. In the first one to two weeks, some people report temporary fatigue, mild joint achiness, or subtle brain fog, and the usual explanation offered is that clearance flush.

One caution about that framing, because it matters. If feeling better proves the protocol is working and feeling worse also proves it is working, the claim cannot be tested, and an untestable claim is a poor reason to keep paying. So I treat worsening as information, never as a milestone. Sometimes it is the flush and it passes. Sometimes it means the dose is too high, the lymphatic and hydration groundwork in Key 6 was skipped, or this is the wrong approach for you. Significant worsening is a reason to reassess with me, not a badge to wear.

The rest of the sequence is structured the same way: weeks three and four are where the early adjustment usually settles, weeks five through eight are where the antioxidant and mitochondrial work is most active, and weeks nine through twelve are the deepest phase of the same work. That is a description of what the protocol is doing in each window. It is not a forecast of how you will feel in each window, and I am not going to give you one.

Setting expectations, at an educational level. Take a baseline before you change anything: energy out of ten, sleep out of ten, and how you feel in the two days after a training session. Then judge everything against your own log rather than against anyone's promise, including mine. Symptoms that are severe, that are not improving after the first couple of weeks, or that appear or worsen later in the process (week four or beyond) are the signal to stop guessing and talk to a practitioner, since later-onset symptoms are handled differently than the expected early adjustment window. And if twelve weeks of your own log shows nothing moved, that is a real result. Stop paying for it.

KEY 8

This Runs Alongside Your Medication, Not Against It

The hesitation worth addressing. It is a reasonable question: if GLP-1 medications cause this cellular side effect, does addressing it mean working against your prescription? No. And understanding why is worth knowing, because it resolves a real, and reasonable, hesitation.

Two pathways, one medication. GLP-1 receptor activity itself has documented anti-aging potential at the receptor signaling level, including activation of AMPK and SIRT1 (pathways linked to cellular longevity), upregulation of NRF2 (a master regulator of your antioxidant defenses), promotion of autophagy, and suppression of NF-kB, a key inflammatory signaling pathway.[12] In other words, the same receptor your medication is designed to activate has real protective effects. The cellular aging consequence described throughout this guide is a separate, tissue-specific effect: GLP-1 receptor stimulation specifically on adipose-derived stem cells, compounded by the sheer speed of fat mobilization outpacing your mitochondria's capacity to process it. These two things, the receptor's genuine benefits and the fat-mobilization side effect, are happening at the same time, in different tissues, through different pathways. That is the paradox, and it is exactly why this guide exists: to address the specific downside without touching the benefit you are already getting.

The bottom line, at an educational level. Keep taking your GLP-1 medication exactly as your prescriber directs. Nothing in this guide is designed to replace, adjust, or interfere with that prescription. The interventions covered here (senolytic support, antioxidant rebuilding, lymphatic clearance, protein and resistance training) are additive. They run concurrently, targeting a documented side effect of the process your medication has already set in motion, so that the version of you on the other side of this weight loss is stronger and healthier, not just lighter.

The 90-Day GLP-1 Cellular Restoration Program

Everything in this guide describes the problem and the general categories of solution. The GLP-1 Cellular Restoration Program, which I built at Seay Wellness, is the structured, personalized version: a 90-day protocol that addresses all four mechanisms covered in this guide at once, senescent cell clearance, antioxidant restoration, physical clearance of cellular debris, and structural rebuild of lean mass, rather than leaving you to piece it together on your own.

The program is built on a core-and-enhanced structure. Every patient starts with a core protocol on day one, covering the foundational support that every GLP-1 patient benefits from regardless of individual lab results: omega-3s, magnesium, vitamin D, collagen peptides, lymphatic drainage support, and senolytic compounds on the correct pulsed schedule. Then, after a telehealth visit in week two where your actual lab results are reviewed, an enhanced layer is added, personalized to what your bloodwork shows you actually need. This is the “most expensive supplement is the one you do not need” principle from Key 5, put into practice instead of just explained.

Two telehealth consultations with me are built into every tier at no extra cost: an intake call that screens your full medication list for interactions before anything ships, and a week-two call where your labs are reviewed and your personalized protocol is finalized.

Foundation: the complete core-and-enhanced supplement protocol, shipped nationally to any US address, with your full patient protocol guide covering dosing, diet, and the resistance exercise protocol from Key 2. No in-office visits required.

Complete: everything in Foundation, plus in-office sessions at Seay Wellness in Garner, NC, combining photobiomodulation laser therapy, pneumatic lymphatic compression, infrared sauna, and neurological recovery support, along with a functional assessment (grip strength, posture, range of motion) at intake and again at the end, so you are comparing yourself against your own starting numbers rather than against a description from memory.

Elite: everything in Complete, plus a full lab panel at intake and again at the end, including biological age testing. A word on that one, because it is oversold everywhere else: biological age tests are an estimate built from a model, not a measurement of your cells, and the models disagree with each other. What it gives you is a consistent number you can run twice and compare to itself. That is genuinely useful as a before-and-after on your own record. It is not proof of anything, and I will not present it to you as proof.

If you have made it this far, you already understand more about what rapid GLP-1 driven weight loss is doing to your cells than most people ever will. The next step is finding out exactly where you stand.

Take the Next Step: Your Free Root Discovery Session

The Root Discovery Session is a complimentary consultation with my team to review your specific situation: how long you have been on your medication, what symptoms you are noticing, and which tier of the program fits where you are right now. There is no obligation, and if you decide to move forward, this conversation is simply the first step.

Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery.

You can also reach the office directly at (919) 662-0520.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, start your free tracker at rootcauseunlocked.com/tracker and take the baseline described in Key 7 before you change anything.

Seay Wellness | 503 US-70 East, Suite L, Garner, NC 27529 rootcauseunlocked.com

This guide is provided for general educational purposes and does not constitute individualized medical advice, diagnosis, or treatment. Always consult your physician before starting any new supplement or exercise program, particularly if you are pregnant, nursing, or taking prescription medications. Continue your GLP-1 medication exactly as prescribed by your physician.

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Every citation below was independently retrieved and confirmed against its live PubMed record (esearch plus esummary or efetch) before use. None are drawn on trust from any prior document. Full verification detail, including claims investigated and left uncited, is in the companion file GLP1-Citation-Ledger.md.

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